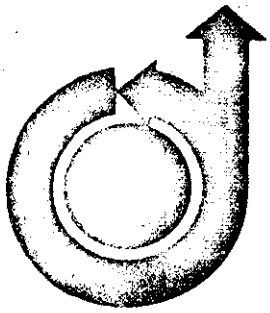
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NASA 30,000 Hour Test Demonstration of Closed Brayton Cycle Reliability

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NASA 30,000 HOUR TEST DEMONSTRATION OF CLOSED BRAYTON CYCLE RELIABILITY

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Abstract

Four Brayton Rotating Units (BRU) were developed by the AiResearch Manufacturing Company of Arizona and, in 1968, were shipped to the NASA Lewis Research Center for performance and endurance testing. The purpose of the tests was to demonstrate the technology for space applications of the closed Brayton power conversion systems. Two of these closed Brayton cycle gas turbine engines (Units 2 and 4) have accumulated over 32,000 hours of successful operation while achieving cycle efficiencies of 30 to 33 percent (References 1 and 2). Both units met or exceeded performance objectives with no detectable performance degradation during endurance testing. Unit 2 was disassembled after 20,000 hours of operation, and given a thorough post-test inspection. The inspection revealed that there was no apparent wear or failure mode that would prevent attainment of the 5-year life objective, and longer life capability seemed probable (Reference 3). Units 1 and 3 were used in early development testing, refurbished, and held in standby. They were not needed to support the endurance test because of the exceptionally high reliability of Units 2 and 4.

The advanced state of closed Brayton cycle technology is evidenced by the fact that the BRUs were designed, built, and shipped to the NASA without the benefit of any prior development testing by AiResearch, e.g., none of the units was ever operated at temperature before delivery. The BRU program has been a dramatic demonstration of design skill, contractor confidence in the product, and maturity of the technology.

Discussion

The BRU is shown in Figure 1. The rotating assembly (Figure 2) operates at a speed of 36,000 rpm and consists of a radial outflow compressor, four-pole Rice alternator/motor and radial inflow turbine. These components are mounted on a common shaft supported on a gas bearing system. The cycle working fluid (inert gas mixture of helium and xenon, molecular weight of 83.8) serves as a medium for bearing lubrication and cooling. The BRU produces from 2.25 to 10.5 kW_e of 1200 Hz, three phase electrical power; design life is 5-10 years at 1600°F turbine inlet temperature. In 1972 an upgraded version of the BRU was delivered to the NASA. Designated BRU-F, this machine develops 15 kWe and utilizes a hydrodynamic, compliant foil gas bearing system.

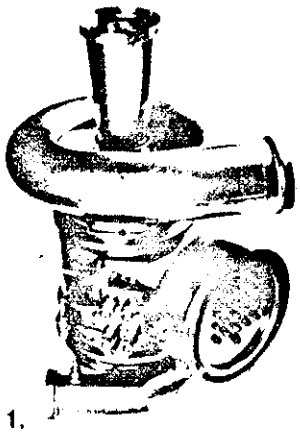


FIGURE 1. NASA BRAYTON ROTATING UNIT (BRU)

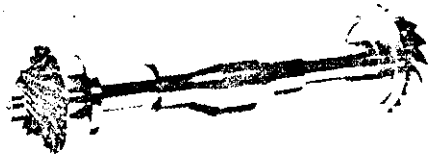


FIGURE 2. BRU ROTATING GROUP

BRUs have been tested by the NASA both as a component and, extensively, as part of a complete flight-type engine system. The power conversion loop, consisting of the rotating unit, recuperator/cooler module, electric heat source, and ducting and bellows was used to completely evaluate BRU performance. Start-stop, thermal transient, design point, off design, and endurance tests have been conducted in both hard vacuum and earth atmospheric environments.

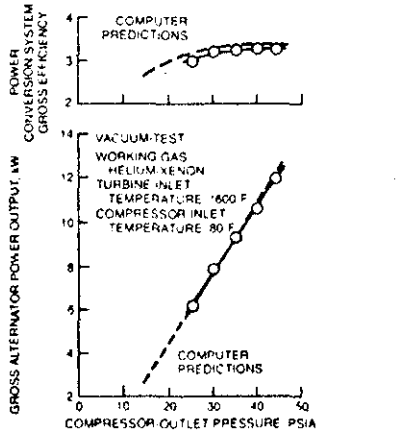
BRU-2 was tested for 20,568 hours with no measurable performance degradation and highest demonstrated cycle efficiency of 33 percent, as shown in Figure 3 (reprinted from Reference 1). A teardown inspection after 20,568 hours revealed that the unit was in such excellent condition that it was reassembled with the original parts, placed on endurance test, and is presently operating unattended. BRU-2 testing is summarized below:

- A 1000-hour design and off-design component performance test was performed.

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- o A 2000-hour test was run as part of a complete engine system at the NASA Plumbrook space vacuum facility.

- o A 17,568-hour endurance test was conducted at 1600°F turbine inlet temperature, with the unit producing 7 to 8 kW_e. Included in the test were 64 start-stop cycles and 16 thermal cycles (cooling to room temperature).



COMPRESSOR OUTLET PRESSURE EFFECT ON POWER CONVERSION SYSTEM GROSS PERFORMANCE DESIGN OPERATING CONDITIONS

FIGURE 3. BRU PERFORMANCE

The BRU-4 engine was tested in parallel with the BRU-2 as part of the performance and endurance test program. An inspection following a scheduled shutdown after 5730 hours revealed a small heat-sink heat-exchanger leak. This low temperature component was repaired, and the loop was operated for an additional 5487 hours, for a total accumulation of 11,217 hours. The unit performed normally throughout this test. This phase of the test program was then terminated, and the test facility and loop dismantled. Teardown inspection of BRU-4 indicated all critical components were in the same excellent condition as observed after 5730 hours.

The endurance tests and post-test inspections verified the mechanical integrity of the design. The only measurable creep detected, occurring in the turbine wheel of BRU-2, was insignificant at less than 0.03 percent in 20,000 hours. Alternator insulation was physically unchanged, and performance unaffected by 20,000 hours of operation; the gas-bearing system was demonstrated to be extremely rugged and reliable. The only potential life-limiting component was the turbine scroll, identified as such after 20,000 hours. This weakness has been eliminated in the design of newer generation machines by changing the configuration from the torus, as used in the BRU, to a simple plenum. The plenum design is easier to fabricate, eliminates most of the present torus welds, and has increased wall thickness, thus reducing the operating stresses and increasing cycle fatigue life. Little, if any, performance compromise has been demonstrated with the plenum configuration.

The significance of the NASA BRU testing may be summarized by the following two observations:

- o There was no performance degradation for two units in over 20,000 hours of endurance operation.

- o After 20,000 hours of operation, there was no apparent wear or failure mode to prevent attainment of the 5-year BRU design life objective.

Conclusion

AiResearch is confident that 7 to 10 year closed Brayton cycle engine life is achievable. This confidence is derived from an intimate knowledge of the system failure modes and effects and design stress margin data.

Currently, AiResearch is developing a "Mini-BRU" space power system designed to produce from 500 to 2500 W_e for 7 years of unattended operation in space. Success of this ERDA-sponsored Brayton Isotope Power System (BIPS) program has, as a basis, the reliability demonstrated by the BRU program. While proven BRU design characteristics have been retained, the NASA BRU development tests have led to improved Mini-BRU design features during the past 5 years which further enhances the high reliability and long life of the BIPS.

The NASA experience, to date, with closed Brayton cycle power conversion is a landmark event in the development history of dynamic power systems for space. The wisdom of the NASA decision in the early 1960's that the closed Brayton cycle represented the greatest potential for this function has been confirmed. The claim that "closed Brayton technology readiness has been achieved" is now justified.

References

1. J. L. Klann, et al, "Performance of the Electrically Heated 2 to 15 kW_e Brayton Power System," NASA TMX-52824, 1970.

2. A. S. Valerino and L. W. Ream, "Performance of a Brayton Cycle Power Conversion System Using Helium-Xenon Gas Mixture," NASA TMX-67846, 1971

3. J. H. Dunn, "Inspection of Two BRUs After Extensive Endurance Testing," NASA TMX-73569, 1977.

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